

Capstone Project Report

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TRIPLE POWER LAW FIT ANALYSIS RESULTS

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DATA STATISTICS:

Total data points: 628

logMacc range: -10.469 to -4.777

logAge range: 5.010 to 7.780

Mstar range: 0.149 to 10.037 Solar Masses

CORRELATIONS:

logAge vs logMacc: -0.150

logMstar vs logMacc: 0.537

logAge vs logMstar: 0.381

TRIPLE POWER LAW FIT RESULTS:

$\log(\text{Macc}) = -4.359 + -0.585 \times \log(\text{Age}) + 1.875 \times \log(\text{Mstar})$

$\text{Macc} = 4.372\text{e-}05 \times \text{Age}^{-0.585} \times \text{Mstar}^{1.875}$

INTERPRETATION:

→ Mass accretion decreases with age (expected)

→ More massive stars have higher accretion rates (expected)

PROBABILISTIC ANALYSIS:

Individual c values distribution: $c \sim N(\mu=-4.359, \sigma=0.591)$

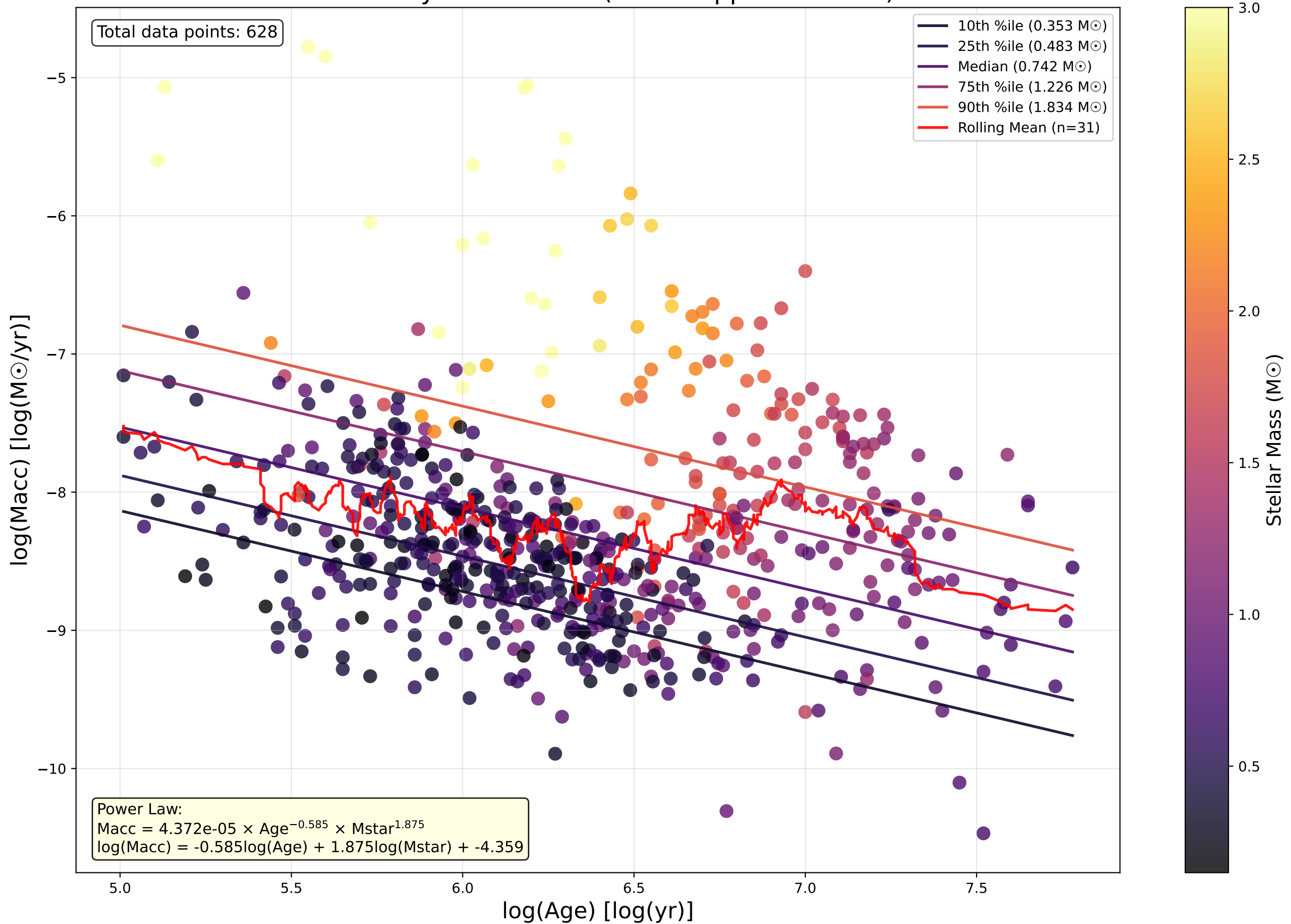
→ Intrinsic scatter in accretion physics: $\sigma = 0.591$

→ 68% of stars have c values within ± 0.591 of mean

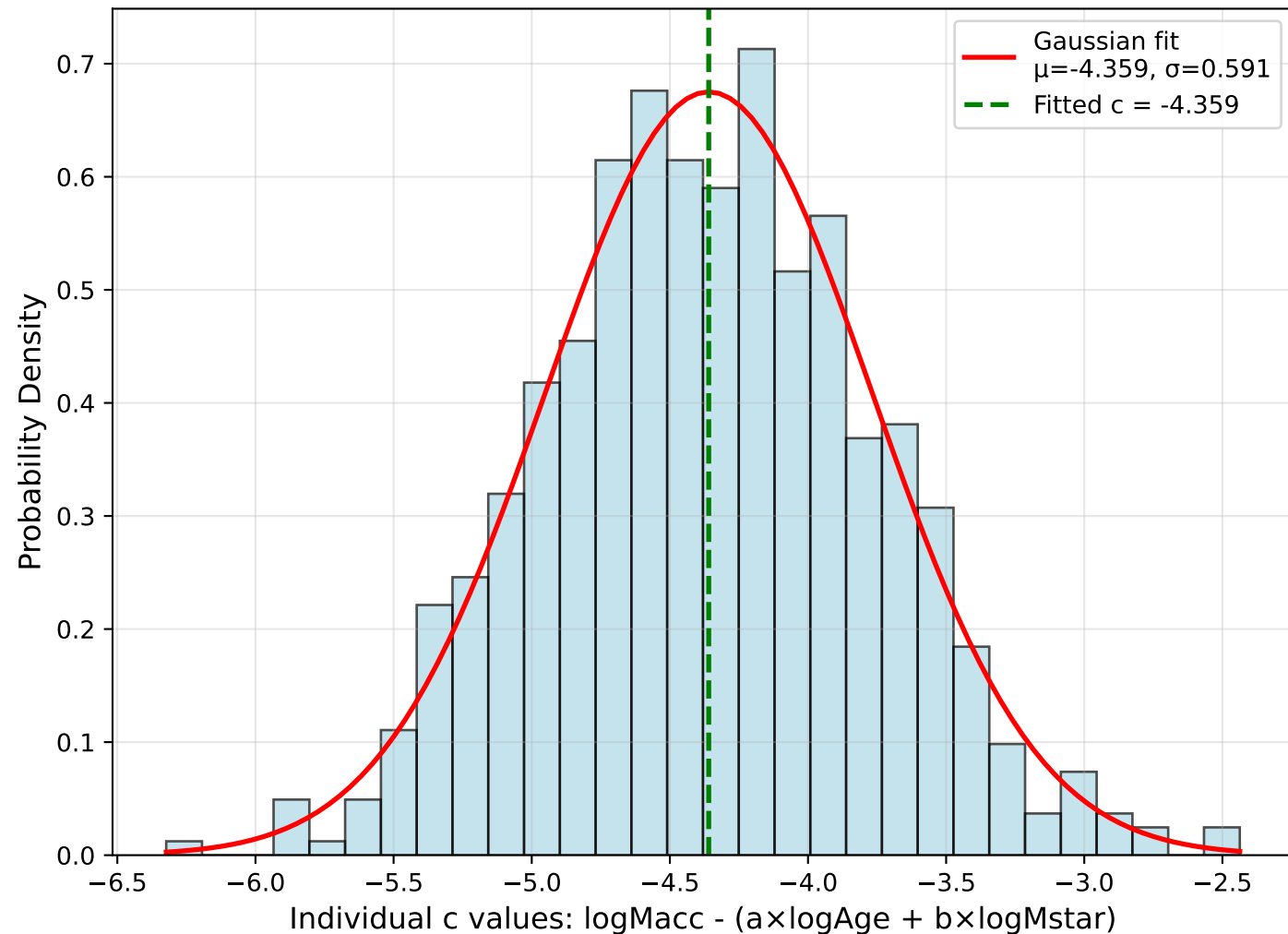
→ Age & mass explain 43.508% of variance

→ Unexplained variance: 56.403%

Mass Accretion Rate vs Age: Triple Power Law Fit
Colored by Stellar Mass (scale capped at 3 M $\odot$)



Distribution of Possible c Values
(Each star has its own "c" value)



Individual c Values vs $\log(\text{Age})$
Colored by Stellar Mass (scale capped at $3 M_{\odot}$)

